Math 4261: Mathematical Statistics I

Lecture 15: Bayesian inference I

(References: Rice 8.5–8.6; DS 7.2–7.4)

Fall 2026

Outline

Finishing up Fisher information

- Asymptotic efficiency
- Fisher information for multiple parameters

Bayesian approach

- A motivating example
- Bayesian inference

Recap: Confidence intervals

An interval $\widehat{I} = \widehat{I}(X_1, \ldots, X_n)$ is an asymptotic $(1-\alpha)$-confidence interval for θ if

$$\mathbb{P}_\theta[\theta \in \widehat{I}] \longrightarrow 1-\alpha \qquad \text{as } n \to \infty.$$

Suppose $\widehat{\theta}$ is consistent and asymptotically normal:

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, v(\theta)),$$

where v is continuous and positive at θ. Then we can use

$$\widehat{I} = \left[\widehat{\theta} - z^{(\alpha/2)}\sqrt{\frac{v(\widehat{\theta})}{n}}, \quad \widehat{\theta} + z^{(\alpha/2)}\sqrt{\frac{v(\widehat{\theta})}{n}}\right],$$

where $z^{(\alpha/2)}$ is the upper-$\alpha/2$ point of the standard Normal.

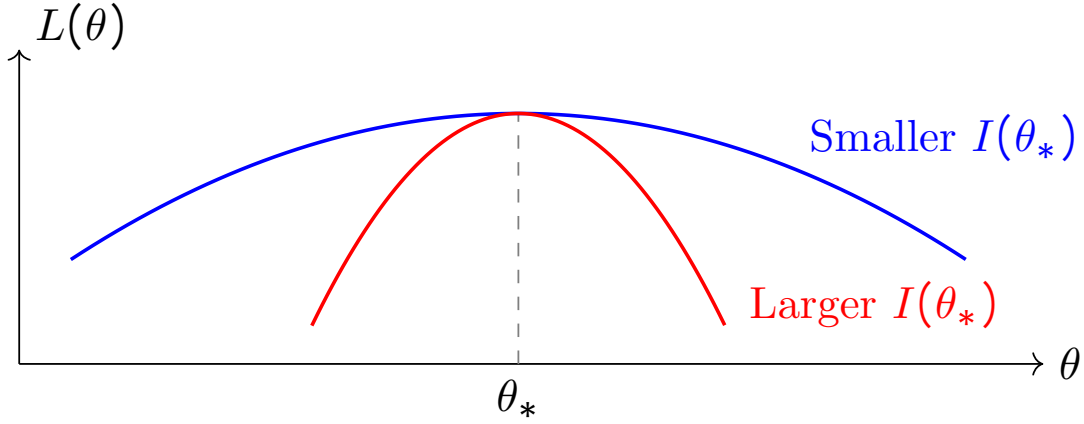
When using the MLE $\widehat{\theta}$, put $v(\theta) = 1/I(\theta)$.

Recap: Fisher information as curvature

In MLE, we maximize $\ell_n(\theta)$, equivalently $\ell_n(\theta)/n$. By the LLN, under regularity conditions, this is approximately maximizing the expected log-likelihood

$$L(\theta) = \mathbb{E}_{\theta_*}[\log f(X \mid \theta)].$$

Fisher information $I(\theta_*) = -L''(\theta_*)$ measures the curvature of $L(\theta)$ at the true parameter θ_*.



Larger Fisher information means a sharper peak and a smaller MLE Normal-approximation variance $1/[nI(\theta_*)]$.

Recap: Cramer-Rao lower bound

Theorem

Under regularity conditions, let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ where θ is a single parameter. Let $T = T(X_1, \ldots, X_n)$ be any unbiased estimator of θ. Then

$$\text{Var}_\theta[T] \geq \frac{1}{nI(\theta)}.$$

- No unbiased estimator can have variance below this bound, regardless of how much computation or ingenuity we use in creating it.
- The proof is based on the correlation between T and the sample score

$$Z = \ell_n'(\theta) = \sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(X_i \mid \theta).$$

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Asymptotic efficiency

An estimator $\widehat{\theta}$ is **asymptotically efficient** if

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta)}\right).$$

That is, the variance of the Normal approximation to $\widehat{\theta}$ for large n attains the Cramer-Rao lower bound of $\frac{1}{nI(\theta)}$. Our main theorem thus shows that the MLE is asymptotically efficient.

If two estimators $\widehat{\theta}_1$ and $\widehat{\theta}_2$ based on $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ satisfy

$$\sqrt{n}(\widehat{\theta}_1 - \theta) \xrightarrow{d} \mathcal{N}(0, v_1(\theta))$$
$$\sqrt{n}(\widehat{\theta}_2 - \theta) \xrightarrow{d} \mathcal{N}(0, v_2(\theta))$$

where $v_2(\theta) < v_1(\theta)$, then $v_2(\theta)/v_1(\theta)$ is the **asymptotic relative efficiency** of $\widehat{\theta}_1$ relative to $\widehat{\theta}_2$.

Asymptotic relative efficiency

Since the Normal approximations have variances $\frac{v_1(\theta)}{n}$ and $\frac{v_2(\theta)}{n}$ for large n, relative efficiency may also be interpreted as the **ratio of sample sizes** needed for $\widehat{\theta}_1$ and $\widehat{\theta}_2$ to achieve the same variance:

If $v_2(\theta) = 80\% \times v_1(\theta)$, then $\widehat{\theta}_1$ using n samples achieves roughly the same variance as $\widehat{\theta}_2$ using only $80\% \times n$ samples.

Example: (Revisited). For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Pareto}(\theta, 1)$, for $\theta > 2$, we showed that

$$\sqrt{n}(\widehat{\theta}_{\text{MoM}} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta(\theta-1)^2}{\theta-2}\right)$$

$$\sqrt{n}(\widehat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}\left(0, \theta^2\right)$$

Thus, the asymptotic relative efficiency of the MoM to MLE is $\frac{\theta(\theta-2)}{(\theta-1)^2}$ which is < 1.

Cramer-Rao for plug-in estimators

The Cramer-Rao lower bound also holds for unbiased estimators of $g(\theta)$.

For the MLE $\widehat{\theta}$, the plug-in estimate $g(\widehat{\theta})$ for $g(\theta)$ satisfies, by the delta method,

$$\sqrt{n}(g(\widehat{\theta}) - g(\theta)) \xrightarrow{d} \mathcal{N}\left(0, \frac{g'(\theta)^2}{I(\theta)}\right).$$

Thus the Normal approximation has variance $\frac{g'(\theta)^2}{nI(\theta)}$ for large n.

Theorem

Under regularity conditions, let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$ where θ is a single parameter. Let $T = T(X_1, \ldots, X_n)$ be any unbiased estimator of $g(\theta)$. Then

$$\text{Var}_\theta[T] \geq \frac{g'(\theta)^2}{nI(\theta)}.$$

Bottom line: efficiency of MLE carries over to smooth functions.

Example: Plug-in estimator for Pareto

Recall that for Pareto$(\theta, 1)$, we found the MLE $\widehat{\theta} = \frac{n}{\sum_{i=1}^{n} \log X_i}$.

The plug-in estimate $\frac{\widehat{\theta}}{\widehat{\theta}-1}$ is not the only reasonable estimate of the mean of the Pareto distribution. What if we decided to simply use the **sample mean** $\overline{X}$?

For this estimate $\overline{X}$, the CLT gives

$$\sqrt{n}\left(\overline{X} - \frac{\theta}{\theta - 1}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta}{(\theta-1)^2(\theta-2)}\right)$$

where $\frac{\theta}{(\theta-1)^2(\theta-2)}$ is the variance of the Pareto distribution assuming $\theta > 2$.

Example: Plug-in estimator for Pareto

For $\theta > 2$, comparing the plug-in and sample mean estimators:

$$\sqrt{n}\left(\frac{\widehat{\theta}}{\widehat{\theta}-1} - \frac{\theta}{\theta-1}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta^2}{(\theta-1)^4}\right)$$

$$\sqrt{n}\left(\overline{X} - \frac{\theta}{\theta-1}\right) \xrightarrow{d} \mathcal{N}\left(0, \frac{\theta}{(\theta-1)^2(\theta-2)}\right)$$

Check: the asymptotic variance corresponding to $\overline{X}$ is **larger** than that of the plug-in! So for large n, the plug-in estimate using the MLE is more accurate than the sample mean $\overline{X}$.

Intuition: Pareto distribution is *heavy-tailed*, and the sample mean $\overline{X}$ is heavily influenced by rare but large data values. In contrast, the MLE $\widehat{\theta}$ estimates θ using $\log X_i$, and then estimates the mean from its relationship to θ.

Potential downside: The plug-in approach relies strongly on the correctness of the Pareto model. On the other hand $\overline{X}$ would be a reasonable estimate of the mean even if Pareto model isn't truly appropriate.

Fisher information for multiple parameters

For a parametric model with p parameters $\theta \in \Omega \subseteq \mathbb{R}^p$, the **Fisher information matrix** $I(\theta) \in \mathbb{R}^{p \times p}$ is the matrix whose (i, j) entry is defined by the equivalent expressions

$$\begin{aligned} I(\theta)_{i,j} &= \operatorname{Cov}_\theta \left[\frac{\partial}{\partial \theta_i} \log f(X \mid \theta), \frac{\partial}{\partial \theta_j} \log f(X \mid \theta) \right] \\ &= -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(X \mid \theta) \right]. \end{aligned}$$

For $p = 1$, the 1×1 matrix $I(\theta)$ recovers our previous definition for Fisher information of a single parameter.

Asymptotic normality of the MLE

The inverse of this Fisher information matrix describes the asymptotic covariance matrix of the MLE $\widehat{\theta} \in \mathbb{R}^p$.

Theorem

Let data $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$, where $\theta \in \Omega \subseteq \mathbb{R}^p$ is the parameter vector. Under regularity assumptions

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1}).$$

The limiting distribution is a p-dimensional multivariate normal distribution whose covariance is the $p \times p$ matrix inverse $I(\theta)^{-1}$.

Example: Gamma MLE

(Revisited). Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Gamma}(\alpha, \lambda)$, which has PDF

$$f(x \mid \alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0, \ \alpha, \lambda > 0.$$

$$\log f(x \mid \alpha, \lambda) = \alpha \log \lambda - \log \Gamma(\alpha) + (\alpha - 1) \log x - \lambda x.$$

$$\frac{\partial}{\partial \alpha} \log f(x \mid \alpha, \lambda) = \log \lambda - \frac{\partial}{\partial \alpha} \log \Gamma(\alpha) + \log x$$

$= \psi(\alpha)$ digamma function

$$\frac{\partial}{\partial \lambda} \log f(x \mid \alpha, \lambda) = \frac{\alpha}{\lambda} - x$$

Example: Gamma MLE

$$\frac{\partial^2}{\partial \alpha^2} \log f(x|\alpha,\lambda) = -\frac{\partial^2}{\partial \alpha^2} \log \Gamma(\alpha)$$

$\psi_1(\alpha)$ trigamma function

$$\frac{\partial^2}{\partial \alpha \partial \lambda} \log f(x|\alpha,\lambda) = \frac{1}{\lambda} \quad , \quad \frac{\partial^2}{\partial \lambda^2} \log f(x|\alpha,\lambda) = -\frac{\alpha}{\lambda^2}$$

So

$$I(\alpha,\lambda) = -\mathbb{E}_{\alpha,\lambda}\begin{bmatrix} \frac{\partial^2}{\partial \alpha^2} \log f(x|\alpha,\lambda) & \frac{\partial^2}{\partial \alpha \partial \lambda} \log f(x|\alpha,\lambda) \\ \frac{\partial^2}{\partial \alpha \partial \lambda} \log f(x|\alpha,\lambda) & \frac{\partial^2}{\partial \lambda^2} \log f(x|\alpha,\lambda) \end{bmatrix}$$

$$= \begin{bmatrix} \psi_1(\alpha) & -\frac{1}{\lambda} \\ -\frac{1}{\lambda} & \alpha/\lambda^2 \end{bmatrix}$$

Example: Gamma MLE

$$\frac{1}{\psi_1(\alpha)\frac{\alpha}{\lambda^2} - \frac{1}{\lambda^2}} \begin{bmatrix} \alpha/\lambda^2 & 1/\lambda \\ 1/\lambda & \psi_1(\alpha) \end{bmatrix}$$

Informally, for large n, the MLEs are approximately distributed as

$$\begin{bmatrix} \widehat{\alpha} \\ \widehat{\lambda} \end{bmatrix} \approx \mathcal{N}\left(\begin{bmatrix} \alpha \\ \lambda \end{bmatrix}, \frac{1}{n} I(\alpha, \lambda)^{-1} \right)$$

Observation: the approximate covariance of $\widehat{\alpha}$ and $\widehat{\lambda}$ is the off-diagonal entry of $\frac{1}{n} I(\alpha, \lambda)^{-1}$, which is

$$\frac{1}{n} \frac{\lambda}{\psi_1(\alpha)\alpha - 1}$$

This can be checked to be always positive, implying that the errors $\widehat{\alpha} - \alpha$ and $\widehat{\lambda} - \lambda$ are positively correlated for large n.

Outline

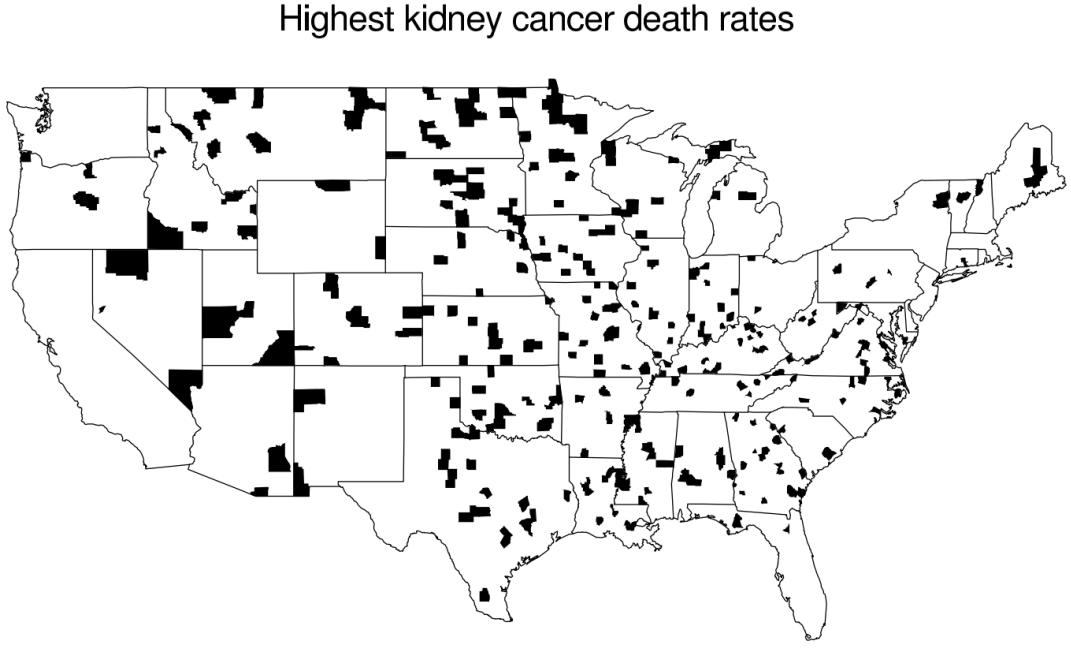
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Example: Kidney cancer in US

Counties of the US with the **highest** 10% age-standardized death rates for cancer of kidney/ureter for US white males 1980-1989.

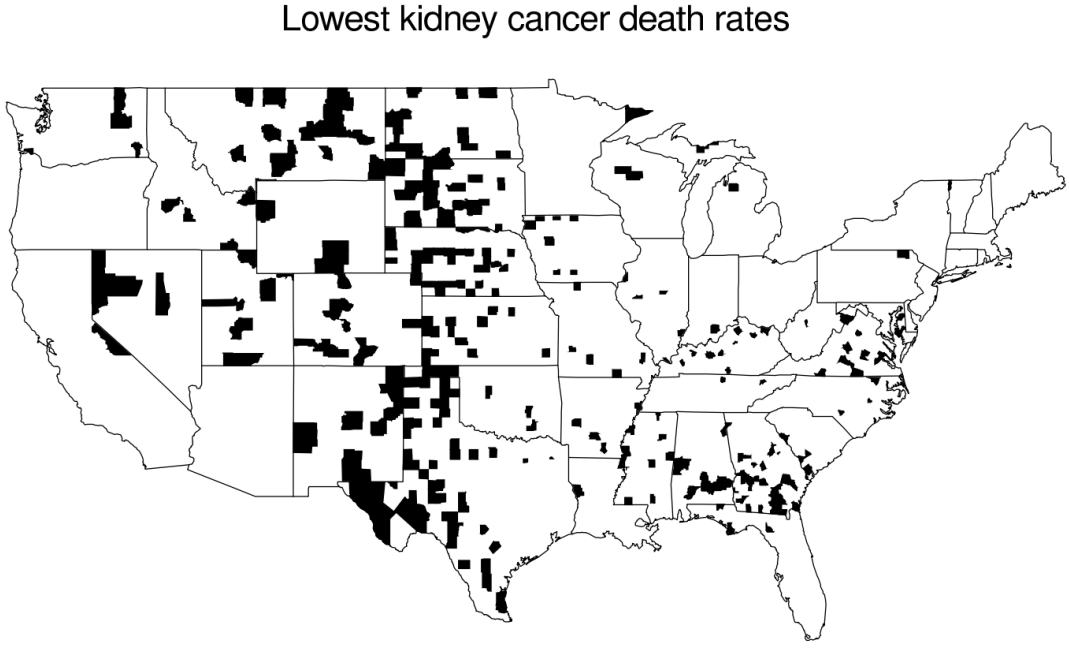
Why are most of the shaded counties in the middle of the country?



BDA3–Gelman et al. and Manton et al.(1989)

Example: Kidney cancer in US

Counties of the US with the **lowest** 10% age-standardized death rates for cancer of kidney/ureter for US white males 1980-1989.



BDA3–Gelman et al. and Manton et al.(1989)

Example: Kidney cancer in US

This is an issue of sample sizes.

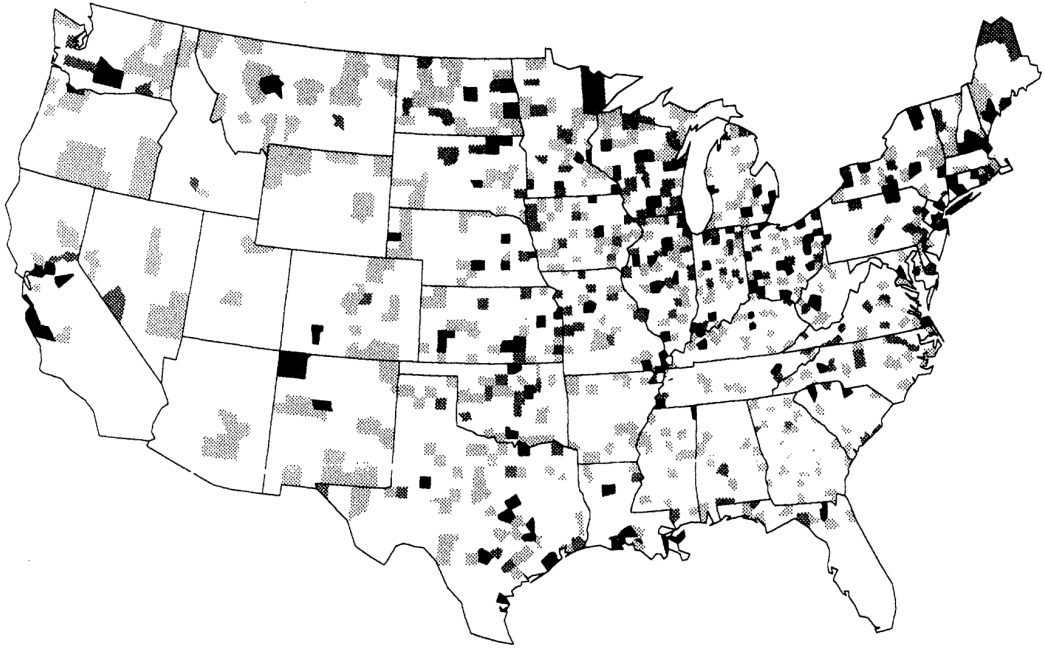
- Kidney cancer is rare
- In any ten-year period, a county of 1000 will probably have zero kidney cancer deaths, so tied for lowest rate in the country and shaded in second picture.
- However, there is a chance the county will have one kidney cancer death during the decade. If so, it will have rate 1 per 10000 per year. High enough to put it in top 10% so shaded in first picture.
- Great Plains has many low-population counties, so overrepresented in the maps.

Bayesian methods provide principled ways to:

- Adjust for county size and rarities in our estimates
- Incorporate *prior* information (e.g. national cancer rates) in our estimates.

Example: Kidney cancer in US

Three levels of shading indicating the counties with the highest 25%, 10%, and 5% **adjusted** age-standardized death rates for cancer of kidney/ureter for US white males over a decade.



BDA3–Gelman et al. and Manton et al.(1989)

Bayesian worldview

Two approaches to parameter estimation

- Method of moments estimator
- Maximum likelihood estimation

Previous discussion has treated θ as an unknown but deterministic parameter, which describes the distribution of our data. This is called the **frequentist** paradigm.

Today we introduce the **Bayesian** paradigm, where Θ is also modeled as random. Bayesian inference starts with our prior belief about Θ, and updates this belief based on observed data.

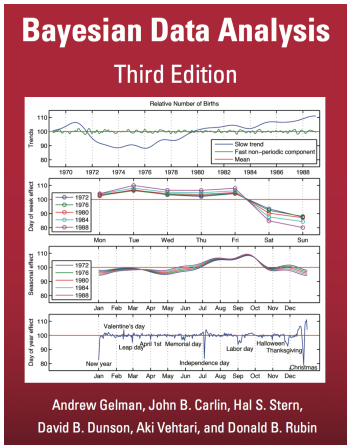
Bayesian worldview

'God created the world in seven days. and we haven't seen much of him since.'

Andrew Gelman (Columbia University)

(God draws Θ from an urn and then is out of the picture).





Recap: Joint-Marginal-Conditional

For continuous[1] X and Y.

Joint PDF is $f_{X,Y}$ is used as

$$\mathbb{P}[(X,Y) \in A] = \iint_A f_{X,Y}(x,y)\,\mathrm{d}x\mathrm{d}y$$

Marginal PDF of X:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}y$$

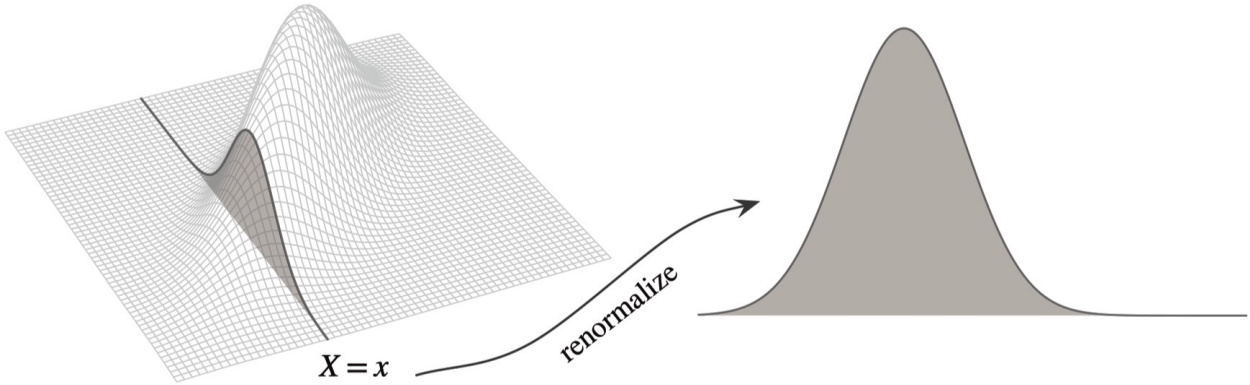
Conditional PDF of Y given $X = x$:

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{\text{Joint PDF}}{\text{Marginal PDF}}$$

[1] replace integrals as sums appropriately for discrete X and Y

Recap: Joint-Marginal-Conditional

Conditional PMF/PDF tells us how to update the distribution of Y, having observed that $X = x$.



Prior and Likelihood

In Bayesian inference, the unknown parameter Θ is modeled as a random variable, with a probability distribution

$$\Theta \sim f_\Theta(\theta)$$

called its **prior distribution**. This represents the experimenter/statistician/scientist's prior belief about the value of Θ, before observing any data.

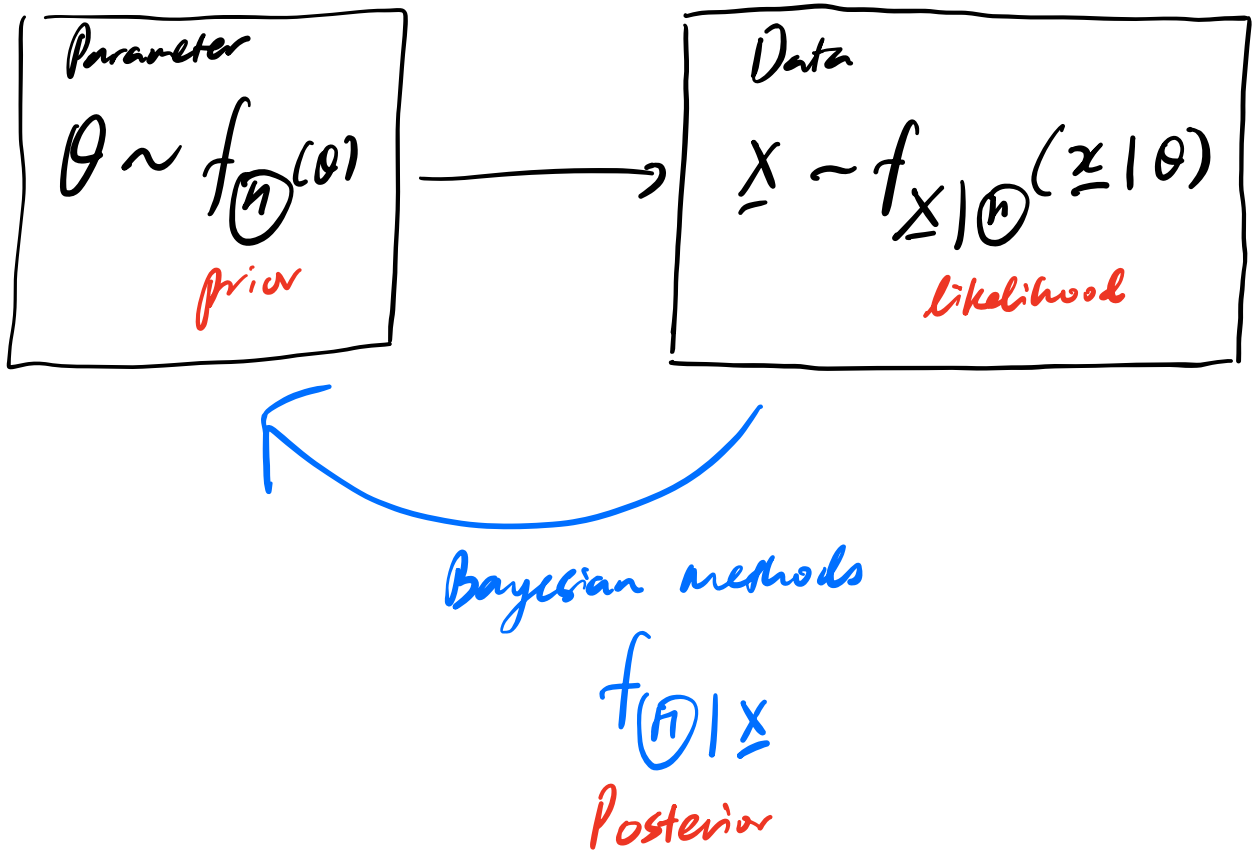
The parametric model or **likelihood function** describing the distribution of our data $\boldsymbol{X} = (X_1, \ldots, X_n)$ is then interpreted as a conditional distribution of $\boldsymbol{X}$ given Θ, which we now write as

$$\boldsymbol{X} \sim f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta).$$

(In previous lectures, we wrote this simply as $f(\boldsymbol{x} \mid \theta)$.)

Prior and Likelihood

Generative process of the data:



Marginal distribution of the data

Conceptually: Prior represents knowledge of the parameter Θ before seeing data.

Mathematically: Θ is just another random variable.

Parameter Θ and observed data $\boldsymbol{X}$ have a joint distribution (PDF/PMF)

$$f_{\boldsymbol{X},\Theta}(\boldsymbol{x}, \theta) = f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta) f_{\Theta}(\theta).$$

Under this joint distribution, the marginal distribution of Θ is simply the prior $f_{\Theta}(\theta)$. The marginal distribution of the data $\boldsymbol{X}$ is

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \int f_{\boldsymbol{X},\Theta}(\boldsymbol{x}, \theta) \, \mathrm{d}\theta = \int f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta) f_{\Theta}(\theta) \, \mathrm{d}\theta$$

This represents the prob. distribution for our observed data $\boldsymbol{X}$, if we assume Θ is also random and marginalize over its randomness.

Posterior distribution

Bayesian inference is based upon the conditional distribution of the parameter Θ given the data $\boldsymbol{X}$. This conditional distribution is

$$f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x}) = \frac{f_{\boldsymbol{X},\Theta}(\boldsymbol{x},\theta)}{f_{\boldsymbol{X}}(\boldsymbol{x})} = \frac{f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta) f_{\Theta}(\theta)}{f_{\boldsymbol{X}}(\boldsymbol{x})}$$

This is called the **Posterior distribution** of Θ. It represents our knowledge about the parameter Θ after observing $\boldsymbol{X}$.

The preceding equation is often summarized simply as

$$f_{\Theta|\boldsymbol{X}}(\theta \mid \boldsymbol{x}) \propto f_{\boldsymbol{X}|\Theta}(\boldsymbol{x} \mid \theta) \cdot f_{\Theta}(\theta)$$
$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

The symbol $\propto$ hides the proportionality factor $1/f_{\boldsymbol{X}}(\boldsymbol{x})$ which does not depend on θ.

Example: Uniform prior for Bernoulli

Let $X_1, \ldots, X_n \sim \text{Bern}(p)$ i.i.d. The parameter p is unknown and is modeled as random with a prior $P \sim \text{Unif}(0,1)$. Derive the posterior.

prior $f_P(p) = 1$ for $p \in (0,1)$

$$f_{X_i|P}(x_i|p) = \begin{cases} p & \text{if } x_i = 1 \\ 1-p & \text{if } x_i = 0 \end{cases} = p^{x_i}(1-p)^{1-x_i}$$

joint $f_{\underline{X},P}(x_1, \ldots, x_n, p) = \underbrace{\left(\prod_{i=1}^{n} f_{X_i|P}(x_i|p)\right)}_{\text{likelihood}} \underbrace{f_P(p)}_{\text{prior}}$

$$= p^{\sum_{i=1}^{n} x_i = s}(1-p)^{n - \sum_{i=1}^{n} x_i}$$

$$= p^s (1-p)^{n-s}$$

Example: Uniform prior for Bernoulli

Marginal distrib. of $\underline{X} = (X_1, \ldots, X_n)$ is

$$f_{\underline{X}}(x_1, \ldots, x_n) = \int_0^1 f_{\underline{X}, p}(x_1 \ldots x_n, p)\, dp$$

$$= \int_0^1 p^S (1-p)^{n-S}\, dp = B(S+1, n-S+1)$$

where $B(\alpha, \beta)$ is the Beta function

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)} = \int_0^1 p^{\alpha-1}(1-p)^{\beta-1}\, dp$$

Hence

$$f_{p|\underline{X}}(p \mid x_1, \ldots, x_n) = \frac{f_{\underline{X}, p}(x_1, \ldots, x_n, p)}{f_{\underline{X}}(x_1, \ldots, x_n)} = \frac{1}{B(S+1, n-S+1)} p^S (1-p)^{n-S}$$

i.e. Posterior is $p \mid \underline{X} \sim \text{Beta}(S+1, n-S+1)$, where $S = X_1 + \cdots + X_n$

Example: Uniform–Bernoulli: Laplace boys girls

Estimation of sex ratios within a population of human births.

- Proportion of births that are female has long been a topic of interest both scientifically and to the lay public.
- Two hundred years ago it was established that the proportion of female births in European populations was < 0.5. Currently accepted proportion in large European-race populations is 0.485.
- This century: interest has focused on factors that may influence the sex ratio.

Letting Y be the number of female births in n births. Laplace used the Uniform-Bernoulli model to estimate girl births in a population (births in Paris from 1745 to 1770). Letting p be the probability that any birth is female, he showed the posterior probability

$$\mathbb{P}\left[P \geq 0.5 \mid Y = 241945, n = 251527 + 241945\right] \approx 1.15 \times 10^{-42}$$

Laplace was 'morally certain' that $p < 0.5$.

Tip for computing posteriors

We computed explicitly the marginal distribution $f_{\boldsymbol{X}}(\boldsymbol{x})$, but this was not actually necessary to derive the posterior. Indeed, omitting factor not depending on p,

$$f_{P|\boldsymbol{X}}(p \mid \boldsymbol{x}) \propto f_{\boldsymbol{X}|P}(\boldsymbol{x} \mid p) f_P(p) \propto p^s (1-p)^{n-s}$$

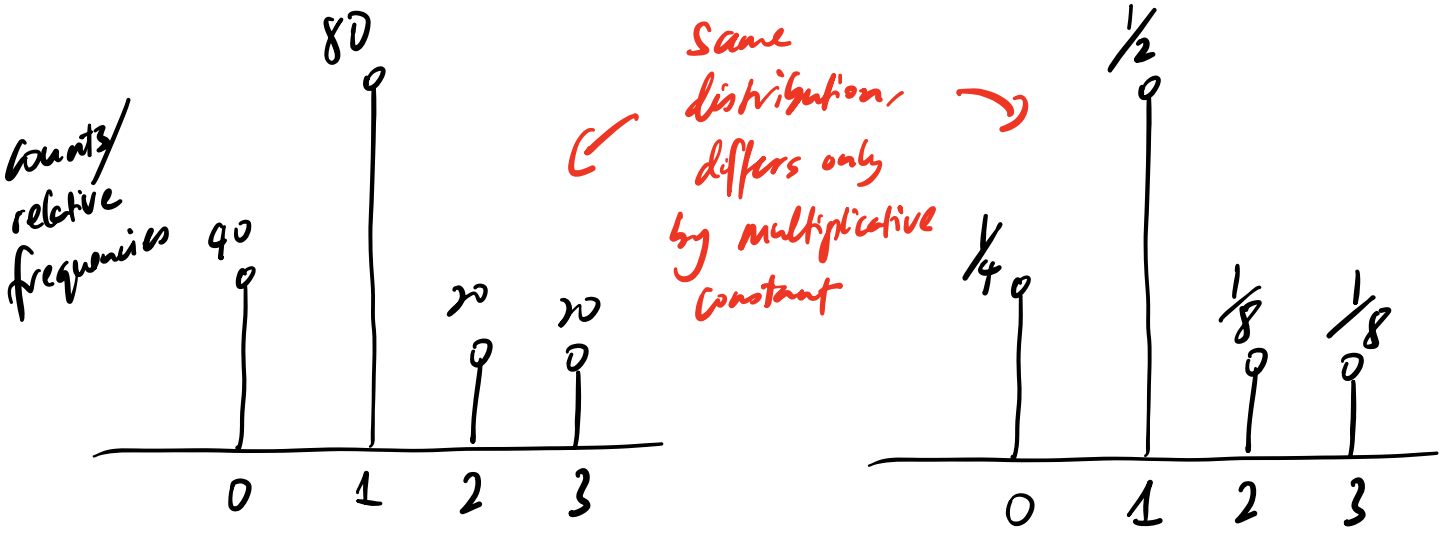
This means the PDF of the posterior distribution of P has the form

$$f_{P|\boldsymbol{X}}(p \mid \boldsymbol{x}) = \frac{1}{C(\boldsymbol{x})} p^s (1-p)^{n-s}$$

where the proportionality constant $C(\boldsymbol{x})$ is the unique number that makes this PDF integrate to 1. Deduce (by probability knowledge) that $C(\boldsymbol{x})$ is the normalizing constant for $\text{Beta}(s+1, n-s+1)$, and that the posterior must be $\text{Beta}(s+1, n-s+1)$.

Tip for computing posteriors: intuition

It is the *shape* of a distribution that determines what it is. Scaling the probabilities by a constant does not change the shape.



E.g. exp(−quadratic function of x) is Gaussian.

Example: Beta prior for Bernoulli

Let $X_1, \ldots, X_n \sim \text{Bern}(p)$ i.i.d. We model p as random with prior $P \sim \text{Beta}(\alpha, \beta)$. The Beta prior means P is a continuous r.v. with

$$\text{prior}: \; f_P(p) = \frac{1}{B(\alpha,\beta)} p^{\alpha-1}(1-p)^{\beta-1}, \quad \text{for } p \in (0,1).$$

Derive the posterior. (Note: Setting $\alpha = 1$, $\beta = 1$ gives Unif(0,1).)

$$f_{P|\underline{X}}(p|x_1 \ldots x_n) \propto f_{\underline{X}|P}(x_1 \ldots x_n|p) \cdot f_P(p)$$

Drop every factor that does not have p in it.

$$\propto p^s (1-p)^{n-s} \cdot p^{\alpha-1}(1-p)^{\beta-1}$$

where $s = x_1 + \cdots + x_n$

$$= p^{s+\alpha-1}(1-p)^{n-s+\beta-1}$$

prop to PDF of Beta$(s+\alpha, n-s+\beta)$

Deduce

Posterior $P|\underline{X} \sim \text{Beta}(s+\alpha, n-s+\beta)$, where $s = x_1 + \cdots + x_n$